

E-commerce customer growth and personalization





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CLASS : B. Sc (Data Science) Sem-VI

PROJECT TITLE : E-commerce Customer Growth Analytics & Personalization

PACKAGE/LANGUAGE : Python (Pandas, Scikit-Learn), SQL (MySQL), Power BI.

PLATFORM : windows10,11





INTRODUCTION PROJECT

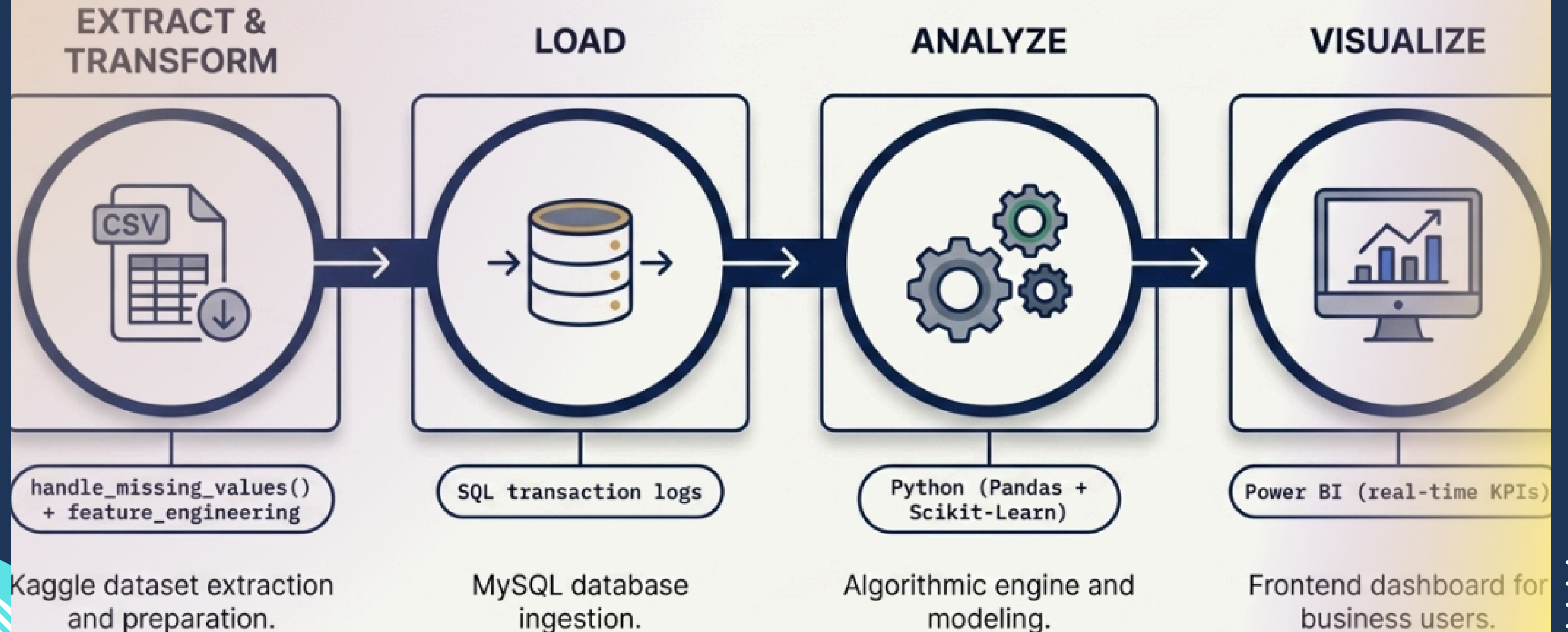
Evaluating An end-to-end data pipeline designed(A data pipeline is structured sequence of step where raw data is processed, cleaned, analyzed, transformed into meaning-full insights) to analyze customer behavior, predict churn(we identify at-risk customer using RFM analysis and predict churn based on inactivity patterns, helping businesses take action to retain them), and provide personalized product recommendations to drive business growth with insights

With the help of RFM and k-means clustering to group the similar data points into clusters.in our project ,we used it to segment customers based on their purchasing behavior using RFM analysis.

The important part of this project, how actually works on industry-level?



THE END-TO-END ANALYTICS PIPELINE



Modules:

- ✓ **ETL(extract,transform, load) & Database Management: Migrating raw e-commerce data into SQL for structured querying.**
Choosing the datasets from olist E-commerce dataset(kaggle)
- ✓ **Exploratory Data Analysis (EDA): Using Python to find trends in sales, geography (e.g., Maharashtra region), and seasonal peaks.**
- ✓ **Customer Segmentation (RFM): Applying K-Means clustering to categorize customers based on value.(e.g.,R:VIP customers,F:Normal customers,M:at-risk)**

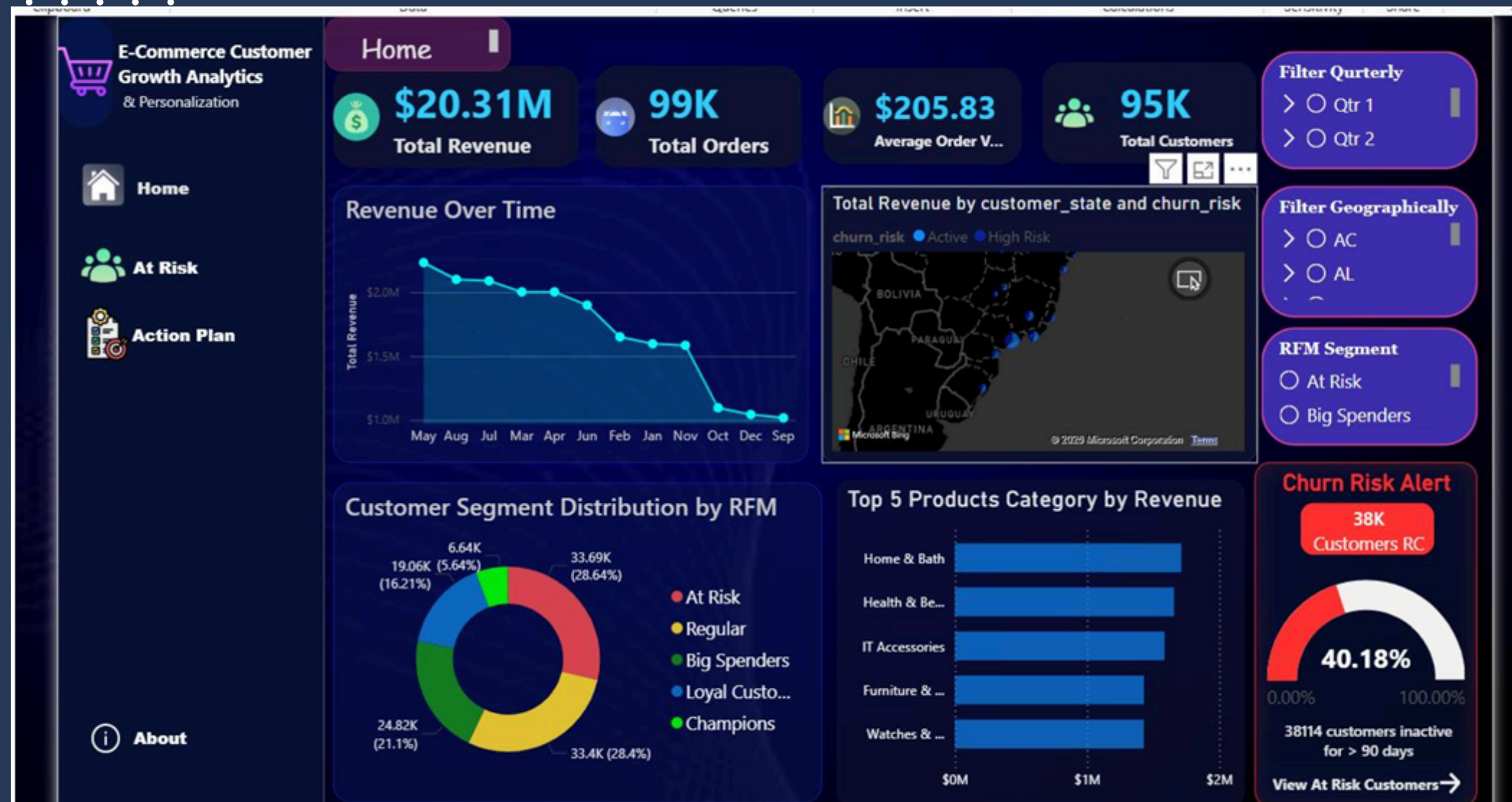


Growth Dashboard: A Power BI interface for real-time monitoring of KPIs(Key Performance Indicator) and retention metrics(repeated customers%).

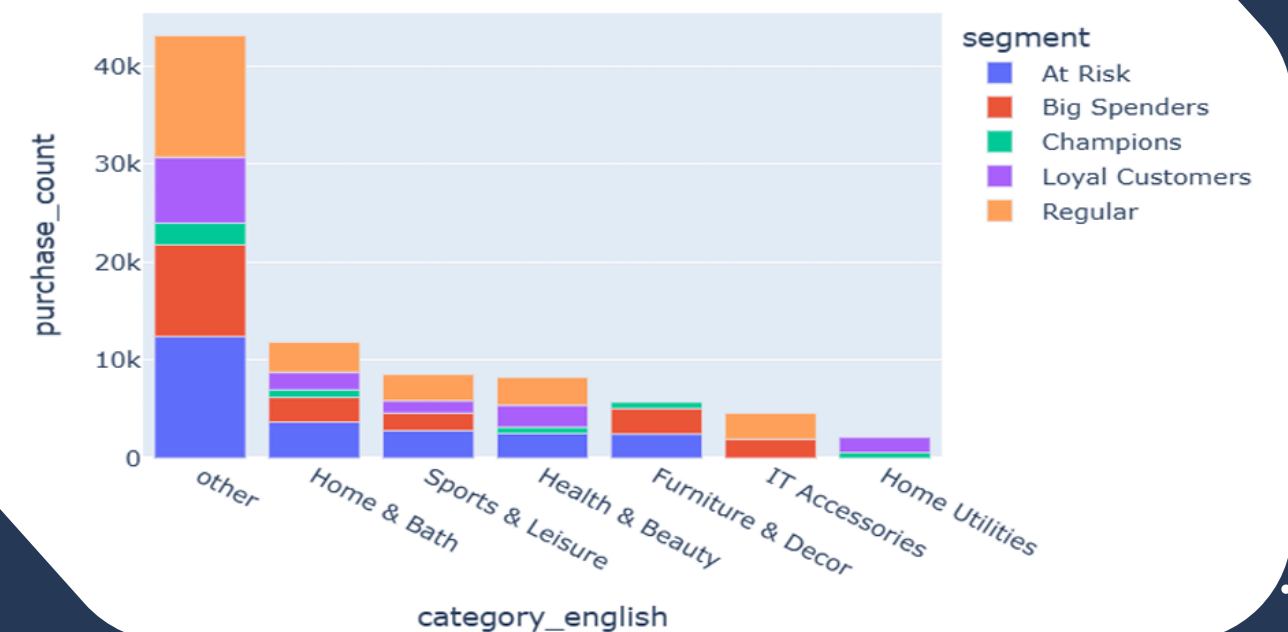
Targeted visuals



Targeted marketing, inventory optimization,
and customer retention strategies for retail
platforms.



Product Categories by Customer Segment





E-Commerce Customer
Growth Analytics
& Personalization



Home



At Risk



Action Plan



About

At Risk



38K

Customers RC



38114 c...

Inactive Custo...



4.11K

Cities At Risk

customer_unique_id	customer_city	churn_risk
09a3f87b74828b274d73ee9dfef43b68	Zortea	Active
d5a5ceb83234eeb855268d7fce7f54ea	Zortea	High Risk
3a96136da5569f486d4f51dbe9cd7246	Ze Doca	Active
9ef06526b8e57a567ed6f1dce7ab0320	Ze Doca	Active
a897335a844c6cb0c0f25a75f43092b4	Ze Doca	High Risk
c933181d09535cb6bdfb1c0160323bd9	Ze Doca	Active
d4798889fe74026a739ca0be9d2dccb8	Ze Doca	High Risk
655fb4f5aca04feecb2f00243473b2c9	Zacarias	High Risk
f43710182f400d86a9aa16fecbbfd6d0	Zacarias	High Risk
39afb6a74f327b36e94ba08376a6029	Xique-Xique	Active
821af681d12f9cd66b20cd1d1e7c02ce	Xique-Xique	Active
d4c0286f064fdfe0879da14087179a69	Xique-Xique	High Risk
27750a2a0de67d552d7ea1cb2892cf1a	Xinguara	High Risk
4261e564d3e674b5234f01d199701475	Xinguara	Active
62d824384528b8f24550482f6f1dd751	Xinguara	Active
64fb97eecb229bb89643a88e5b54d614	Xinguara	Active
8d3bd6b5becadb64bc3adcf4c546d2e5	Xinguara	Active
96767434dc8da40da17994f8011b422c	Xinguara	High Risk
b3fb415f8a02ee0f55ddb4f4ca9df2fe	Xinguara	High Risk
b94783c71788124eb4f2d4e1fcae6ab9	Xinguara	High Risk



**Thank you
so much**